Chp-3

Laplace Transform & Continious-Time LTI systems.

Laplace transform -

$$x(t) \underset{\mathcal{L}^{-1}}{\overset{\mathcal{L}}{\rightleftharpoons}} X(s) \quad ; \; s = \sigma + j\omega \text{ (complex number)}$$

$$Y(s) = X(s) \cdot H(s)$$

$$\mathcal{L}\{x(t)\} = X(s)$$

$$\mathcal{L}^{-1}\{X(s)\} = x(t)$$

$\Rightarrow$

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\, dt$$

$$Y(s) = \mathcal{L}\{y(t)\} = \int_{-\infty}^{\infty} y(t) \cdot e^{-st}\, dt$$

$$H(s) = \mathcal{L}\{h(t)\} = \int_{-\infty}^{\infty} h(t) \cdot e^{-st}\, dt$$

ROC -> Region of convergence

* Pole & zero,

Let, $X(s) = \dfrac{(s+2)}{(s-1)(s+3)}$;

Here, $X(s) = \infty$ for, $s = 1$ & -3 $\therefore$ Pole = $1, -3$

$X(s) = 0$ for, $s = -2$ $\therefore$ zero = -2

Properties of ROC –

1. The ROC doesn't contain any poles.
2. If $x(t)$ is a finite duration signal, then ROC is the entire s-plane except possibly $s=0$ or $s=\infty$.
3. If $x(t)$ is a right sided signal,

$$Re(s) > \sigma_{max}$$

4. If $x(t)$ is a left sided signal,

$$Re(s) < \sigma_{min}$$

5. If $x(t)$ is a both sided signal,

$$\sigma_1 < Re(s) < \sigma_2$$

The inverse Laplace transform method –

1. Inversion formula
2. Laplace tranformation table → Also depend on ROC condition.
3. Partial fraction

↳ Demo

$$\frac{1}{(s+a)(s+b)} = \frac{A}{(s+a)} + \frac{B}{(s+b)}$$

T-1: Laplace transform table

$x(t)$	$X(s)$	ROC
$\delta(t)$	1	All s
$u(t)$	$1/s$	$Re(s) > 0$
$-u(-t)$	$1/s$	$Re(s) < 0$
$t\,u(t)$	$1/s^2$	$Re(s) > 0$
$t^k u(t)$	$\frac{k!}{s^{k+1}}$	$Re(s) > 0$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$Re(s) > -Re(a)$
$-e^{-at}u(-t)$	$\frac{1}{s+a}$	$Re(s) < -Re(a)$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$Re(s) > -Re(a)$
$-te^{-at}u(-t)$	$\frac{1}{(s+a)^2}$	$Re(s) < -Re(a)$
$\cos\omega_0 t \cdot u(t)$	$\frac{s}{s^2+\omega_0^2}$	$Re(s) > 0$
$\sin\omega_0 t \cdot u(t)$	$\frac{\omega_0}{s^2+\omega_0^2}$	$Re(s) > 0$
$e^{-at}\cos\omega_0 t\,u(t)$	$\frac{s+a}{(s+a)^2+\omega_0^2}$	$Re(s) > -Re(a)$
$e^{-at}\sin\omega_0 t\,u(t)$	$\frac{\omega_0}{(s+a)^2+\omega_0^2}$	$Re(s) > -Re(a)$

T-2: Properties of Laplace transform

Property	Signal	Transformation	ROC
	$x(t)$	$X(s)$	R
	$x_1(t)$	$X_1(s)$	R_1
	$x_2(t)$	$X_2(s)$	R_2
Linearity	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(s) + a_2X_2(s)$	$R' \supset R_1 \cap R_2$
Time shifting	$x(t-t_0)$	$e^{-st_0}X(s)$	$R' = R$
Shifting in s	$e^{s_0t}x(t)$	$X(s-s_0)$	$R' = R + Re(s_0)$
Time scaling	$x(at)$	$\frac{1}{\lvert a \rvert}X(s)$	$R' = aR$
Time reversal	$x(-t)$	$X(-s)$	$R' = -R$
Differentiation in t	$\frac{dx(t)}{dt}$	$sX(s)$	$R' \supset R$
Differentiation in s	$-tx(t)$	$\frac{dX(s)}{ds}$	$R' = R$
Integration	$\int_{-\infty}^{t} x(\tau)\,d\tau$	$\frac{1}{s}X(s)$	$R' \supset R \cap \{Re(s) > 0\}$
Convolution	$x_1(t) * x_2(t)$	$X_1(s)\,X_2(s)$	$R' \supset R_1 \cap R_2$

System Function –

Let,

$y(t)$ the output of a continious LTI system and the convolution of the input $x(t)$ with the impulse response $h(t)$.

So, $y(t) = x(t) * h(t)$

$$\Rightarrow Y(s) = X(s) \cdot H(s)$$

$$\Rightarrow H(s) = \frac{Y(s)}{X(s)}$$

The Laplace transform $H(s)$ of $h(t)$ is refered to as the system function of the system.

Ex-3.1

Find Laplace transform -

a) $x(t) = -e^{-at}u(-t)$

b) $x(t) = e^{at}u(-t)$

—o—

a) $x(t) = -e^{-at}u(-t)$

$= e^{-at}u(t)$

$\therefore X(s) = \mathcal{L}\{x(t)\}$

$= \int_{-\infty}^{\infty} e^{-at}u(t)\, e^{-st}\, dt$

$= \int_{-\infty}^{0} e^{-at}u(t)\, e^{-st}\, dt + \int_{0}^{\infty} e^{-at}u(t)\, e^{-st}\, dt$

$= \left[\frac{e^{-t(s+a)}}{-(s+a)}\right]_0^{\infty}$

$= \frac{1}{-(s+a)}\left[e^{-t(s+a)}\right]_0^{\infty}$

$= \frac{1}{s+a}$

$\therefore$ ROC, $Re(s) < -a$

b) $x(t) = e^{-at} u(t)$

$\therefore X(s) = \mathcal{L}\{x(t)\}$

$$= \int_{-\infty}^{\infty} e^{-at} u(t) \cdot e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-at} e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-t(a+s)}\, dt$$

$$= \left[\frac{e^{-t(s+a)}}{-(s+a)} \right]_0^{\infty}$$

$$= -\frac{1}{s+a} \left[e^{-t(s+a)} \right]_0^{\infty}$$

$$= -\frac{1}{s+a} (0 - 1)$$

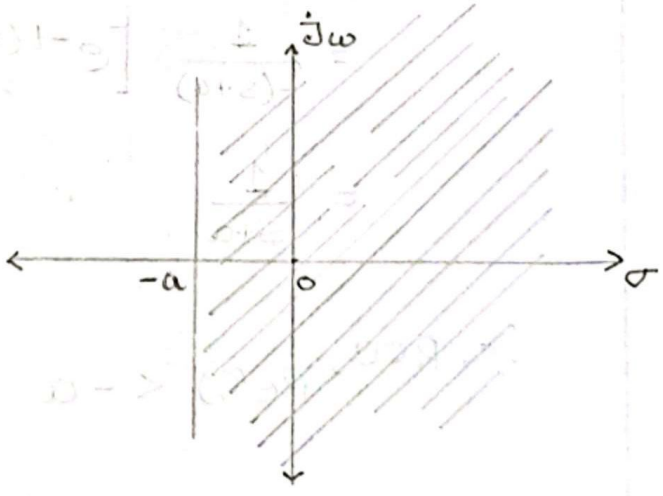
$$= \frac{1}{s+a}$$

$\therefore X(s) = \frac{1}{s+a}$

$\therefore$ ROC,

$Re(s) > -a$

Ans



Ex-3.5

Find the Laplace transform $X(s)$ and sketch the pole-zero plot with the ROC for the signal $x(t)$:

a) $x(t) = e^{-2t}u(t) + e^{2t}u(-t)$

b) $x(t) = e^{-3t}u(t) + e^{2t}u(-t)$

c) $x(t) = e^{2t}u(t) + e^{-3t}u(-t)$

—o—

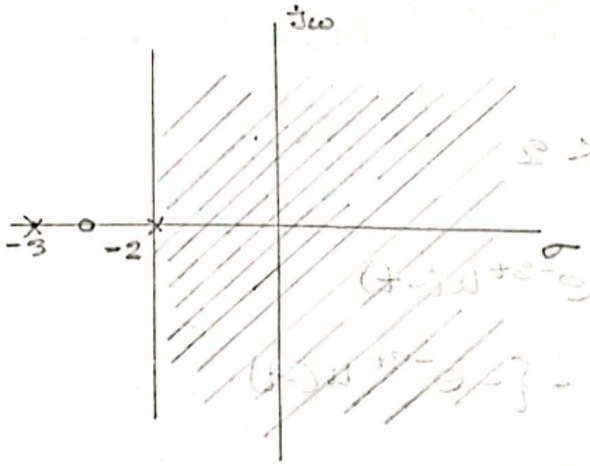
a) $x(t) = e^{-2t}u(t) + e^{-3t}u(t)$

$\therefore X(s) = \frac{1}{s+2} + \frac{1}{s+3}$

For, $\frac{1}{s+2}$, $Re(s) > -2$

$\frac{1}{s+3}$, $Re(s) > -3$

$\therefore Re(s) > -2$



$Re(s) > -2$

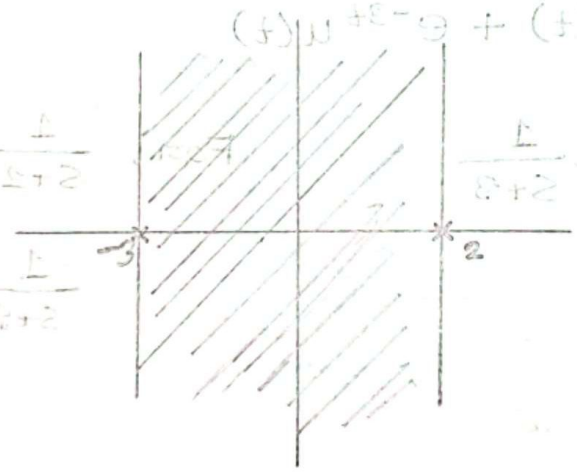
b) $x(t) = e^{-3t}u(t) + e^{2t}u(-t)$

$= e^{-3t}u(t) - \{-e^{(-2)t}u(-t)\}$

$\therefore X(s) = \frac{1}{s+3} - \frac{1}{s-2}$

For, $\frac{1}{s+3}$, $Re(s) > -3$ → For right side signal

$\frac{1}{s-2} = \frac{1}{s+(-2)}$, $Re(s) < -(-2) \Rightarrow Re(s) < 2$ → For left side signal



$\therefore -3 < Re(s) < 2$

c) $x(t) = e^{2t}u(t) + e^{-3t}u(-t)$

$= e^{-(-2)t}u(t) - \{-e^{-3t}u(-t)$

$\therefore X(s) = \frac{1}{s-2} - \frac{1}{s+3}$

$\frac{1}{s-2}$, $Re(s) > 2$

$\frac{1}{s+3}$, $Re(s) < -3$

$\therefore$ 2 < Re(s) < -3

As the signal don't have any common ROC, there will be no graph.

Ex-3.16

Find inverse Laplace transform of the following $X(s)$ -

a) $X(s) = \frac{1}{s+1}$, $Re(s) > -1$

b) $X(s) = \frac{1}{s+1}$, $Re(s) < -1$

c) $X(s) = \frac{s}{s^2+1}$, $Re(s) > 0$

d) $X(s) = \frac{s+1}{(s+1)^2+1}$, $Re(s) > -1$

Using Laplace transform table.

$\Rightarrow$

a) $X(s) = \frac{1}{s+1}$

$\therefore x(t) = \mathcal{L}^{-1}\{X(s)\}$

$= e^{-1 \cdot t} u(t)$

$= e^{-t} u(t)$

$\frac{1}{s+a} = e^{-at} u(t)$; $a = 1$

$Re(s) > -1$; Right sided

b) $X(s) = \frac{1}{s+1}$

$\therefore x(t) = \mathcal{L}^{-1}\{X(s)\}$

$= e^{-t} u(-t)$

$a = 1$

$Re(s) < -1$; Left sided.

c) $X(s) = \frac{s}{s^2+4}$

$\therefore x(t) = \mathcal{L}^{-1}\{X(s)\}$

$= \mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\}$

$= \cos 2t\, u(t)$

$\frac{s}{s^2+\omega_0^2}$; $\omega_0 = 2$

$Re(s) > 0$,

$\frac{s}{s^2+\omega_0^2} = \cos\omega_0 t\, u(t)$

d) $X(s) = \frac{s+1}{(s+1)^2+4}$

$\therefore x(t) = \mathcal{L}^{-1}\{X(s)\}$

$= \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+4}\right\}$

$= e^{-1\cdot t}\cos 2t\, u(t)$

$= e^{-t}\cos 2t\, u(t)$

$\frac{s+a}{(s+a)^2+\omega_0^2} = e^{-at}\cos\omega_0 t\, u(t)$

$\therefore a = 1$; $\omega_0 = 2$

$Re(s) > -1$

Ex-3.17

Find the inverse Laplace transform of the following X(s):

a) $X(s) = \frac{2s+4}{s^2+4s+3}$, $Re(s) > -1$

b) $X(s) = \frac{2s+4}{s^2+4s+3}$, $Re(s) < -3$

c) $X(s) = \frac{2s+4}{s^2+4s+3}$, $-3 < Re(s) < -1$

Method-3

$$\Rightarrow X(s) = \frac{2s+4}{s^2+4s+3} = \frac{2(s+2)}{s^2+3s+s+3} = \frac{2(s+2)}{s(s+3)+1(s+3)}$$

$$\Rightarrow X(s) = \frac{2(s+2)}{(s+3)(s+1)}$$

$$\Rightarrow \frac{X(s)}{2} = \frac{(s+2)}{(s+3)(s+1)} \equiv \frac{A}{(s+3)} + \frac{B}{(s+1)} \quad ---(1)$$

From (1),

$$\frac{s+2}{(s+3)(s+1)} \equiv \frac{A}{(s+3)} + \frac{B}{(s+1)}$$

$$\Rightarrow s+2 \equiv A(s+1) + B(s+3) \quad \left[\text{Multiplying } (s+3)(s+1) \text{ in both side.}\right]$$

For,

$s = -1$,

$$-1+2 = A(-1+1) + B(-1+3)$$

$$\Rightarrow 1 = 2B$$

$$\therefore B = \frac{1}{2}$$

$s = -3$,

$$-3+2 = A(-3+1) + B(-3+3)$$

$$\Rightarrow -1 = -2A$$

$$\therefore A = \frac{1}{2}$$

Putting, A & B value in eq-1,

$$\frac{s+2}{(s+3)(s+1)} = \frac{\frac{1}{2}}{s+3} + \frac{\frac{1}{2}}{s+1}$$

$$\Rightarrow \frac{X(s)}{2} = \frac{1}{2(s+3)} + \frac{1}{2(s+1)}$$

$$\therefore X(s) = \frac{1}{s+3} + \frac{1}{s+1} \quad \therefore x(t) = e^{-3t}u(t)$$

a) For ROC, $Re(s) > -1$,

The signal is right sided,

So, $x(t) = e^{-3t}u(t) + e^{-1t}u(t)$

$= e^{-3t}u(t) + e^{-t}u(t)$

$= u(t)\{e^{-3t} + e^{-t}\}$

b) For ROC, $Re(s) < -3$,

The signal is left sided,

So, $x(t) = -e^{-3t}u(-t) - e^{-t}u(-t)$

$= u(-t)\{-e^{-3t} - e^{-t}\}$

c) For, ROC, $-3 < Re(s) < -1$

The signal is both sided,

So, $x(t) = e^{-3t}u(t) - e^{-t}u(-t)$

(Ans)

Ex-3.20

Find the inverse Laplace transform of the following X(s):

a) $X(s) = \frac{2s+1}{s+2}$, $Re(s) > -2$

b) $X(s) = \frac{s^2+6s+7}{s^2+3s+2}$, $Re(s) > -1$

c) $X(s) = \frac{s^3+2s^2+6}{s^2+3s}$, $Re(s) > 0$

—o—

a) $X(s) = \frac{2s+1}{s+2} = \frac{2(s+2)-3}{s+2} = 2 - \frac{3}{s+2}$

As, $ROC_{Re(s)} > -2$, this is a right sided signal,

$\therefore x(t) = 2 \cdot 1 - 3e^{-2t}u(t)$

$= 2\delta(t) - 3e^{-2t}u(t)$

Ans

b) $X(s) = \frac{s^2+6s+7}{s^2+3s+2} = \frac{s^2+3s+2+3s+5}{s^2+3s+2}$

$= \frac{s^2+3s+2}{s^2+3s+2} + \frac{3s+5}{s^2+3s+2}$

$= 1 + \frac{3s+5}{s^2+2s+s+2}$

$= 1 + \frac{3s+5}{s(s+2)+1(s+2)} = 1 + \frac{3s+5}{(s+2)(s+1)}$

Let,

$$X(s) = X_1(s) + X_2(s)$$

$$\therefore X_2(s) = \frac{3s+5}{(s+1)(s+2)} \equiv \frac{A}{(s+1)} + \frac{B}{(s+2)} \quad ---(1)$$

From eq (1),

$$\frac{3s+5}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$\Rightarrow 3s+5 = A(s+2) + B(s+1)$$

For,

$s = -2$,

$$3(-2)+5 = A(-2+2) + B(-2+1)$$

$$\Rightarrow -1 = -B$$

$$\therefore B = 1$$

For,

$s = -1$,

$$3(-1)+5 = A(-1+2) + B(-1+1)$$

$$\Rightarrow 2 = A$$

$$\therefore A = 2$$

$$\therefore \frac{3s+5}{(s+1)(s+2)} = X_2(s) = \frac{2}{s+1} + \frac{1}{s+2}$$

$\therefore X(s) = X_1(s) + X_2(s)$

$$= 1 + \frac{2}{s+1} + \frac{1}{s+2}$$

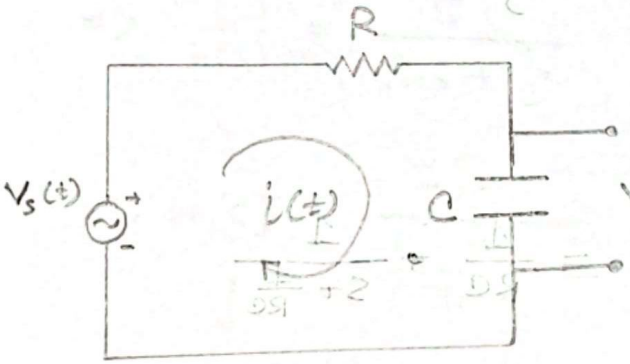
As, ROC; $Re(s) > -1$, this is a right sided signal.

$\therefore x(t) = \delta(t) + 2e^{-t}u(t) + e^{-2t}u(t)$

Ans

Q) ~~$X(s) = \frac{s^3+2s^2+\epsilon}{s^2+3s}$~~

Ex - 3.23



Find the system function H(s) and the impulse response h(t) of the RC circuit.

a) If $x(t) = V_s(t)$ and $y(t) = V_o(t)$

b) If $x(t) = V_s(t)$ and $y(t) = i(t)$

$\Rightarrow$

a) KVL,

$$V_s(t) = V_R(t) + V_c(t)$$

$$\Rightarrow V_s(t) = R \times i(t) + V_c(t)$$

$$\Rightarrow x(t) = R \times i(t) + y(t)$$

$$\Rightarrow x(t) = RC \frac{d}{dt} V_c(t) + y(t)$$

$$\Rightarrow x(t) = RC \frac{d}{dt} y(t) + y(t)$$

$$\Rightarrow X(s) = RC \cdot s \cdot Y(s) + Y(s)$$

[Used table 2 for $\frac{d}{dt} y(t) = s \cdot Y(s)$]

$$\Rightarrow X(s) = Y(s)\{s \cdot RC + 1\}$$

$$\therefore \frac{Y(s)}{X(s)} = \frac{1}{s \cdot RC + 1}$$

(Divided both side by RC)

$$= \frac{\frac{1}{RC}}{s + \frac{1}{RC}} = \frac{1}{RC} \cdot \frac{1}{s + \frac{1}{RC}}$$

$$\therefore H(s) = \frac{Y(s)}{X(s)} = \frac{1}{RC} \cdot \frac{1}{s + \frac{1}{RC}}$$

Used $\frac{1}{s+a}$ from table.

$$\therefore h(t) = \mathcal{L}^{-1}\{H(s)\} = \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$$

Ans

b) KVL,

$$V_s(t) = V_R(t) + V_c(t)$$

$$\Rightarrow V_s(t) = R \times i(t) + V_c(t)$$

$$\Rightarrow x(t) = R\,y(t) + \frac{1}{C}\int i(t)\cdot dt$$

$$\Rightarrow x(t) = R\,y(t) + \frac{1}{C}\int y(t)\,dt$$

$$\Rightarrow \frac{d}{dt}x(t) = \frac{d}{dt}R\cdot y(t) + \frac{1}{C}\frac{d}{dt}\int y(t)\,dt$$

$$\Rightarrow S\cdot X(s) = R\cdot S\cdot Y(s) + \frac{1}{C}Y(s)$$

$$\Rightarrow S\cdot X(s) = Y(s)\left\{R\cdot S + \frac{1}{C}\right\}$$

$$\Rightarrow \frac{Y(s)}{X(s)} = \frac{S}{R\cdot S + \frac{1}{C}}$$

$$\Rightarrow H(s) = \frac{\frac{S}{R}}{\frac{RC\cdot S + 1}{RC}} = \frac{\frac{S}{R}}{\frac{RCS}{RC} + \frac{1}{RC}} = \frac{\frac{S}{R}}{S + \frac{1}{RC}}$$

$$\downarrow$$

$$= \frac{1}{R}\,\frac{S}{S+\frac{1}{RC}}$$

$$= \frac{1}{R}\cdot\frac{S + \frac{1}{RC} - \frac{1}{RC}}{S+\frac{1}{RC}}$$

$$= \frac{1}{R}\cdot\left\{\frac{S+\frac{1}{RC}}{S+\frac{1}{RC}} - \frac{\frac{1}{RC}}{S+\frac{1}{RC}}\right\}$$

P.T.O

$$= \frac{1}{R}\left\{1 - \frac{1}{RC}\frac{1}{s+\frac{1}{RC}}\right\}$$

$$= \frac{1}{R}\left\{1 - \frac{1}{RC}\frac{1}{s+\frac{1}{RC}}\right\}$$

$$= \frac{1}{R} - \frac{1}{R^2C}\frac{1}{s+\frac{1}{RC}}$$

$$\therefore h(t) = \mathcal{L}^{-1}\{H(s)\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{R} - \frac{1}{R^2C}\frac{1}{s+\frac{1}{RC}}\right\}$$

$$= \frac{1}{R} - \frac{1}{R^2C}e^{-\frac{t}{RC}}u(t)$$

Ans

Ex-3.25

The output $y(t)$ of a continuous-time LTI system is found to be $2e^{-3t}u(t)$, when the input $x(t)$ is $u(t)$

a) Find the impulse response $h(t)$ of the system.

b) Find the output $y(t)$ when the input $x(t)$ is $e^{-t}u(t)$.

=>

a) Given,

$$x(t) = u(t)$$

$$y(t) = 2e^{-3t}u(t)$$

$$\therefore X(s) = \frac{1}{s}$$

$$\therefore Y(s) = 2\,\frac{1}{s+3} = \frac{2}{s+3}$$

$$\therefore H(s) = \frac{Y(s)}{X(s)} = \frac{\frac{2}{s+3}}{\frac{1}{s}} = \frac{2s}{s+3} = \frac{2(s+3)-6}{s+3} = \frac{2(s+3)}{s+3} - \frac{6}{s+3}$$

$$= 2 - 6\,\frac{1}{s+3}$$

$$\therefore h(t) = \mathcal{L}^{-1}\{H(s)\} = \mathcal{L}^{-1}\left\{2\cdot 1 - 6\,\frac{1}{s+3}\right\}$$

$$= 2\delta(t) - 6e^{-3t}u(t)$$

Ans

b) Given,

$$h(t) = 2\delta(t) - 6e^{-3t}u(t)$$

$$x(t) = e^{-t}u(t)$$

$$\therefore X(s) = \frac{1}{s+1}$$

$$\therefore H(s) = \frac{2s}{s+3}$$

$$\therefore Y(s) = X(s) \cdot H(s) = \frac{1}{s+1} \cdot \frac{2s}{s+3} = \frac{2s}{(s+1)(s+3)}$$

$$\Rightarrow \frac{Y(s)}{2} = \frac{s}{(s+1)(s+3)} \equiv \frac{A}{s+1} + \frac{B}{s+3}$$

$$\Rightarrow \quad s = A(s+3) + B(s+1)$$

For $s = -1$,

$$-1 = A(-1+3)$$

$$\Rightarrow A = -\frac{1}{2}$$

For $s = -3$,

$$-3 = B(-3+1)$$

$$\Rightarrow B = \frac{3}{2}$$

$$\therefore \frac{Y(s)}{2} = \frac{-\frac{1}{2}}{s+1} + \frac{\frac{3}{2}}{s+3}$$

$$\Rightarrow Y(s) = \frac{-1}{s+1} + \frac{3}{s+3}$$

$$\therefore y(t) = -e^{-t}u(t) + 3e^{-3t}u(t)$$

Ans

Chp-4

Z-Transform and Discrete-Time LTI Systems:

$y[n] = x[n] * h[n]$

$\therefore X(z) = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}$

$x[n] \longleftrightarrow X(z)$

$\therefore Y(z) = \sum_{n=-\infty}^{\infty} y[n] \cdot z^{-n}$

$n \rightarrow z = r\angle\theta$

$\therefore H(z) = \sum_{n=-\infty}^{\infty} h[n] \cdot z^{-n}$

$x[n] = a^n u[n]$ $\therefore X[z] = ?$

$\Rightarrow$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} a^n u[n] \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= 1 + az^{-1} + (az^{-1})^2 + \cdots + (az^{-1})^n$$

$$= \frac{1}{1-az^{-1}} = \frac{1}{1-\frac{a}{z}} = \frac{z}{z-a} \quad \therefore ROC,\ Re(z) > a$$

$r = \frac{az^{-1}}{1} = \frac{(az^{-1})^2}{az^{-1}} = az^{-1}$

$\therefore n$ এর প্রথম সর্তটি,

$\frac{1}{1-r} = \frac{1}{1-az^{-1}}$

Properties of ROC,

1. The ROC does not contain any poles.
2. If $x[n]$ is a finite sequence, then ROC is the the entire z-plane except possibly, $z=0$ or $z=\infty$
3. If $x[n]$ is right sided, sequence,

$$|z| > r_{max}$$

4. If $x[n]$ is left sided, sequence,

$$|z| < r_{min}$$

5. If $x[n]$ is both sided sequence,

$$r_1 < |z| < r_2$$

The inverse z-transform,

A. Inversion formula.

B. Use of Tables of z-Transform Pairs.

C. Power series expansion.

D. Partial-Fraction expansion.

T1: Z-Transform Pairs

$x[n]$	$X(z)$	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{z}{z-1}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\frac{z}{z-1}$	$\lvert z\rvert < 1$
$\delta[n-m]$	z^{-m}	All z except $\begin{cases} 0 ; m>0 \\ \infty ; m<0 \end{cases}$
$a^n u[n]$	$\frac{z}{z-a}$	$\lvert z\rvert > a$
$-a^n u[-n-1]$	$\frac{z}{z-a}$	$\lvert z\rvert < a$
$n a^n u[n]$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert > a$
$-a^n n u[-n-1]$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert < a$
$(n+1) a^n u[n]$	$\left[\frac{z}{z-a}\right]^2$	$\lvert z\rvert > a$
$(\cos \Omega_0 n) u[n]$	$\frac{z^2-(\cos\Omega_0)z}{z^2-(2\cos\Omega_0)z+1}$	$\lvert z\rvert > 1$
$(\sin \Omega_0 n) u[n]$	$\frac{(\sin\Omega_0)z}{z^2-(2\cos\Omega_0)z+1}$	$\lvert z\rvert > 1$
$(r^n \cos \Omega_0 n) u[n]$	$\frac{z^2-(r\cos\Omega_0)z}{z^2-(2r\cos\Omega_0)z+r^2}$	$\lvert z\rvert > r$
$(r^n \sin \Omega_0 n) u[n]$	$\frac{(r\sin\Omega_0)z}{z^2-(2r\cos\Omega_0)z+r^2}$	$\lvert z\rvert > r$
$\begin{cases} a^n ; 0 \le n \le N-1 \\ 0 ; \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$\lvert z\rvert > 0$

T2: Properties of the z-Transform

Property	Sequence	Transform	ROC
	$x[n]$	$X(z)$	R
	$x_1[n]$	$X_1(z)$	R_1
	$x_2[n]$	$X_2(z)$	R_2
Linearity	$a_1x_1[n]+a_2x_2[n]$	$a_1X_1(z)+a_2X_2(z)$	$R' \supset R_1 \cap R_2$
Time shifting	$x[n-n_0]$	$z^{-n_0}X(z)$	$R' \supset R \cap \{0<\lvert z\rvert<\infty\}$
Multiplication by z_0^n	$z_0^n x[n]$	$X\left(\frac{z}{z_0}\right)$	$R' = \lvert z_0\rvert R$
Multiplication by $e^{j\omega_0 n}$	$e^{j\omega_0 n}x[n]$	$X(e^{-j\omega_0}z)$	$R' = R$
Time-reversal	$x[-n]$	$X\left(\frac{1}{z}\right)$	$R' = \frac{1}{R}$
Multiplication by n	$nx[n]$	$-z\frac{dX(z)}{dz}$	$R' = R$
Accumulation	$\sum_{k=-\infty}^{n} x[n]$	$\frac{1}{1-z^{-1}}X(z)$	$R' \supset R \cap \{\lvert z\rvert>1\}$
Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	$R' \supset R_1 \cap R_2$

Ex 4.3

A finite sequence $x[n] = \{5, 3, \underset{\uparrow}{-2}, 0, 4, -3\}$

Find $X(z)$ & ROC.

=>

Given,

$$x[n] = \{5, 3, \underset{\uparrow}{-2}, 0, 4, -3\}$$

$$-2, -1, n=0, 1, 2, 3$$

$$\therefore X(z) = \sum_{n=-2}^{3} x[n]\, z^{-n}$$

$$= x[-2]\, z^{-(-2)} + x[-1]\, z^{-(-1)} + x[0]\, z^{-0} + x[1]\, z^{-1} + x[2]\, z^{-2} + x[3]\, z^{-3}$$

$$= 5z^2 + 3z - 2 + 0 \cdot z^{-1} + 4z^{-2} - 3z^{-3}$$

$$= 5z^2 + 3z - 2 + 4z^{-2} - 3z^{-3}$$

As the signal is finite, ROC is entire z plane except $z = 0$ or $z = \infty$.

Ex 4.6

Find the z-transform X(z) and sketch the pole-zero plot with the ROC -

a) $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n]$

b) $x[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[-n-1]$

c) $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n-1]$

$\Rightarrow$

a) $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n]$

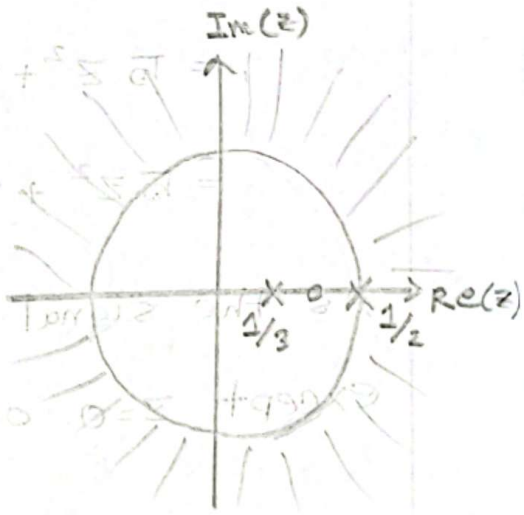
$$\therefore X[z] = \frac{z}{z-\frac{1}{2}} + \frac{z}{z-\frac{1}{3}}$$

$$= \frac{z\left(z-\frac{1}{3}\right) + z\left(z-\frac{1}{2}\right)}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$$

$$= \frac{z^2 - \frac{z}{3} + z^2 - \frac{z}{2}}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$$

$$= \frac{2z^2 - \frac{5}{6}z}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$$

$$= \frac{z\left(2z-\frac{5}{6}\right)}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$$



Poles = $\frac{1}{2}, \frac{1}{3}$

zero = $0, \frac{5}{12}$

$\therefore$ ROC, $|z| > \frac{1}{2}$

b) $x[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[-n-1]$

$$\therefore X(z) = \frac{z}{z-\frac{1}{3}} - \frac{z}{z-\frac{1}{2}}$$

$$= \frac{z\left(z-\frac{1}{2}\right) - z\left(z-\frac{1}{3}\right)}{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{2}\right)}$$

$$= \frac{z^2 - \frac{z}{2} - z^2 + \frac{z}{3}}{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{2}\right)}$$

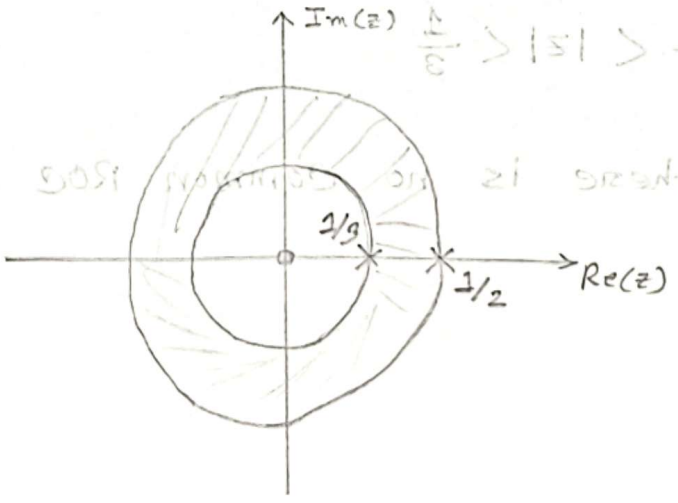
$$= \frac{z\left(-\frac{1}{6}\right)}{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{2}\right)}$$

$$= -\frac{1}{6}\,\frac{z}{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{2}\right)}$$

$\therefore$ ~~Roc~~ poles $= \frac{1}{3}, \frac{1}{2}$

zeros $= 0$

$\therefore$ ROC, $\frac{1}{3} < |z| < \frac{1}{2}$



(a) $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[-n-1]$

$= \left(\frac{1}{2}\right)^n u[n] - \left\{-\left(\frac{1}{3}\right)^n u[-n-1]\right\}$

$\therefore X(z) = \frac{z}{z-\frac{1}{2}} - \frac{z}{z-\frac{1}{3}}$

$= \frac{z\left(z-\frac{1}{3}\right) - z\left(z-\frac{1}{2}\right)}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$

$= \frac{z^2 - \frac{z}{3} - z^2 + \frac{1}{2}}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$

$= \frac{z\left(\frac{1}{6}\right)}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$

$= \frac{1}{6} \frac{z}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}$

$\therefore$ Poles $= \frac{1}{2}, \frac{1}{3}$

Zeros $= 0$

$\therefore$ ROC, $\frac{1}{2} < |z| < \frac{1}{3}$

So here, there is no common ROC exist.

Ex- 4.15

Find the inverse z-transform of

$$X(z) = z^2\left(1-\frac{1}{2}z^{-1}\right)\left(1-z^{-1}\right)\left(1+2z^{-1}\right)$$

=>

Given,

$$X(z) = z^2\left(1-\frac{1}{2}z^{-1}\right)\left(1-z^{-1}\right)\left(1+2z^{-1}\right)$$

$$= \left(z^2-\frac{1}{2}z\right)\left(1-z^{-1}\right)\left(1+2z^{-1}\right)$$

$$= \left(z^2-\frac{1}{2}z\right)\left(1+2z^{-1}-z^{-1}-2z^{-2}\right)$$

$$= \left(z^2-\frac{1}{2}z\right)\left(1+z^{-1}-2z^{-2}\right)$$

$$= z^2+z-2-\frac{1}{2}z-\frac{1}{2}+z^{-1}$$

$$= z^2+\frac{1}{2}z-\frac{5}{2}+z^{-1}$$

$x[n]z^{-n}$

$$= x[-2]\cdot z^{-(-2)} + x[-1]\cdot z^{-(-1)} - z[0]\cdot z^{0} + x[1]z^{-1}$$

$\therefore x[n] = \left\{1, \frac{1}{2}, -\frac{5}{2}, 1\right\}$

↑

Ans

Ex- 4.19

Find the inverse z-transform using partial fraction,

a) $X(z) = \frac{z}{2z^2-3z+1}$; $|z| < \frac{1}{2}$

b) $X(z) = \frac{z}{z-1} - \frac{z}{z-\frac{1}{2}}$; $|z| > 1$

$\Rightarrow$

a) $X(z) = \frac{z}{2z^2-3z+1}$; $|z| < \frac{1}{2}$

$= \frac{z}{2z^2-2z-z+1}$

$= \frac{z}{2z(z-1)-1(z-1)}$

$= \frac{z}{(z-1)(2z-1)}$

$= \frac{z}{2(z-\frac{1}{2})(z-1)}$

$\Rightarrow \frac{X(z)}{z} = \frac{1}{2(z-\frac{1}{2})(z-1)} \equiv \frac{A}{z-\frac{1}{2}} + \frac{B}{z-1}$

$\Rightarrow \quad \Rightarrow \frac{1}{2} = A(z-1) + B\left(z-\frac{1}{2}\right)$

For $z = 1$,

$$\frac{1}{2} = A(1-1) + B\left(1 - \frac{1}{2}\right)$$

$$\Rightarrow \frac{1}{2} = \frac{1}{2}B$$

$$\therefore B = 1$$

For $z = \frac{1}{2}$

$$\frac{1}{2} = A\left(\frac{1}{2} - 1\right) + B\left(\frac{1}{2} - \frac{1}{2}\right)$$

$$\Rightarrow \frac{1}{2} = -\frac{1}{2}A$$

$$\therefore A = -1$$

$$\therefore \frac{X(z)}{z} = \frac{-1}{z - \frac{1}{2}} + \frac{1}{z-1}$$

$$\Rightarrow X(z) = \frac{z}{z-1} - \frac{z}{z - \frac{1}{2}}$$

$$\therefore x[n] = -u[-n-1] - \left\{-\left(\frac{1}{2}\right)^n u[-n-1]\right\}$$

$$= -u[-n-1] + \left(\frac{1}{2}\right)^n u[-n-1]$$

$$= u[-n-1]\left\{\left(\frac{1}{2}\right)^n - 1\right\}$$

Here, $u[-n-1] = \begin{cases} 1 & n \leq -1 \\ 0 & n > 0 \end{cases}$

For,

$n = -1$, $x[-1] = \left(\frac{1}{2}\right)^{-1} - 1 = 2 - 1 = 1$

$n = -2$, $x[-2] = \left(\frac{1}{2}\right)^{-2} - 1 = 4 - 1 = 3$

$n = -3$, $x[-3] = \left(\frac{1}{2}\right)^{-3} - 1 = 8 - 1 = 7$

$n = -4$, $x[-4] = \left(\frac{1}{2}\right)^{-4} - 1 = 16 - 1 = 15$

$\therefore x[n] = \{\ldots 15, 7, 3, 1, 0\}$

Ans

b) $X(z) = \frac{1}{z-1} - \frac{1}{z-\frac{1}{2}}$; $|z| > 1$

$\therefore x[n] = u[n] - \left(\frac{1}{2}\right)^n u[n]$

$= u[n] - \left\{1 - \left(\frac{1}{2}\right)^n\right\}$

Here,

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

For,

$n = 0$, $x[0] = 1 - \left(\frac{1}{2}\right)^0 = 0$

$n = 1$, $x[1] = 1 - \left(\frac{1}{2}\right)^1 = \frac{1}{2}$

$n = 2$, $x[2] = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$

$n = 3$, $x[3] = 1 - \left(\frac{1}{2}\right)^3 = \frac{7}{8}$

$\therefore x[n] = \{0, \frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \ldots\}$